# Predicting Employee Attrition Using Remote Work Productivity and Workplace Factors

GROUP PROJECT

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**Course:** CPSC 5071 – Data Management for Data Science

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# Project Objective

Build a predictive model to identify factors associated with employee attrition by integrating workplace and remote productivity datasets.

The goal is to help organizations better understand attrition patterns and support data-driven retention strategies.

# Problem Statement

## Why Attrition Matters:

- Employee attrition
- High turnover
- Recruitment costs
- Training costs
- Productivity loss
- Job satisfaction
- Overtime workload
- Salary level
- Workplace conditions
- Remote work impact
- Employee productivity

# Data and Data Management

Two datasets from Kaggle were used in this project

Dataset 1: Employee Attrition and Factors
(HR information such as salary, job satisfaction, overtime, and years at company)

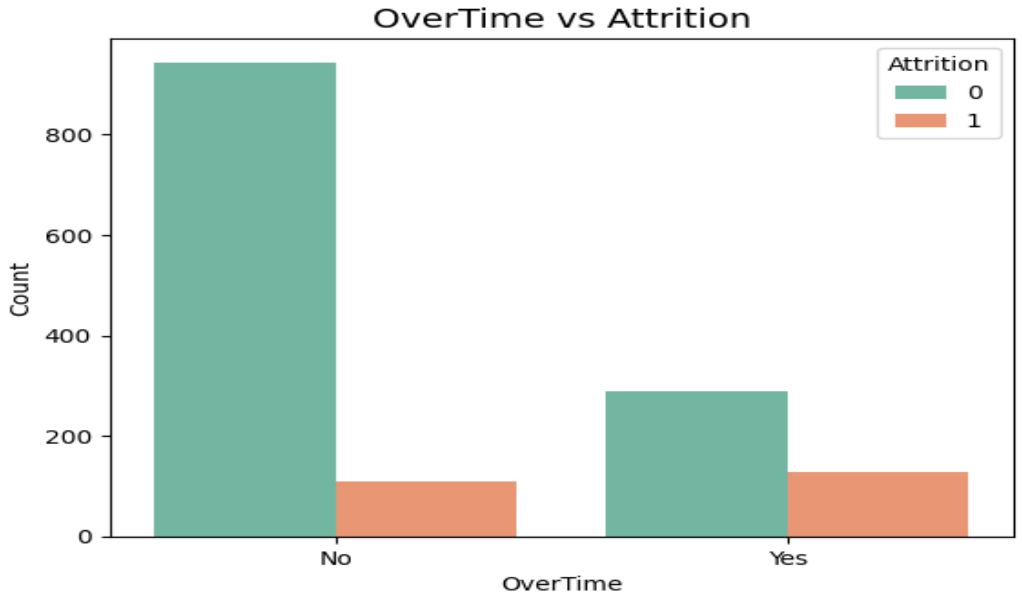
Dataset 2: Remote Worker Productivity Dataset
(productivity, stress level, and work environment factors)

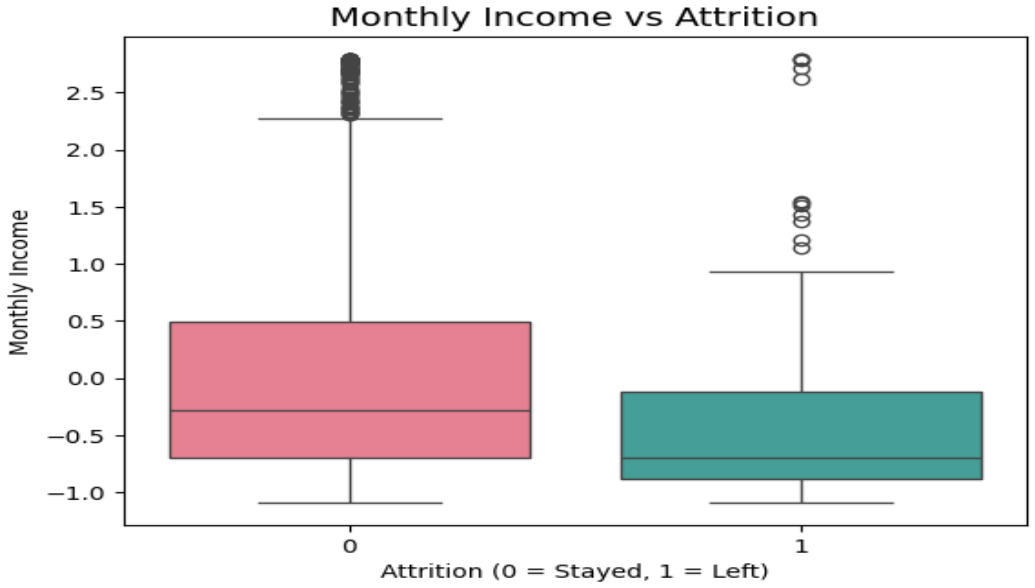
The datasets were integrated based on **experience level** to combine workplace and productivity information

Data preprocessing included cleaning, handling missing values, encoding categorical variables, and preparing the dataset for machine learning models

# Exploratory Data Analysis

- Attrition distribution shows that most employees stay, creating class imbalance.
- Employees who work overtime tend to have a higher likelihood of leaving.
- Lower job satisfaction is associated with higher attrition rates.
- Income and years at the company also show noticeable differences between employees who stay and those who leave.



# Modeling Approach

The dataset was divided using an 80/20 train–test split.

Cross-validation was applied during model training to improve generalization.

Two classification models were implemented:

- Logistic Regression
- Random Forest Classifier

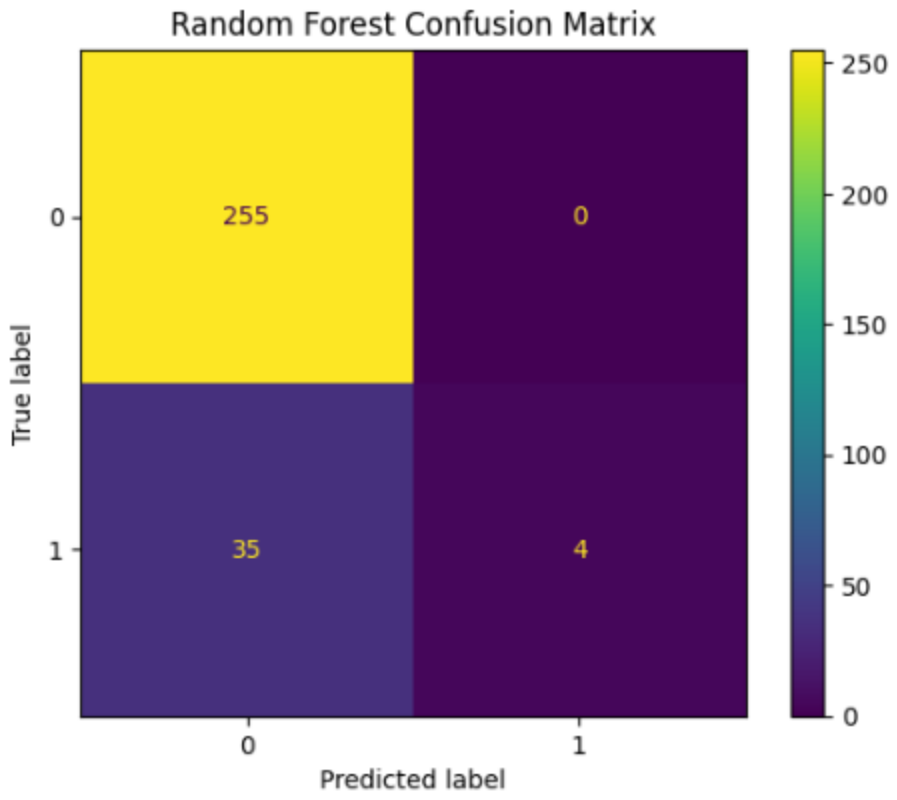
Model performance was evaluated using the following metrics:

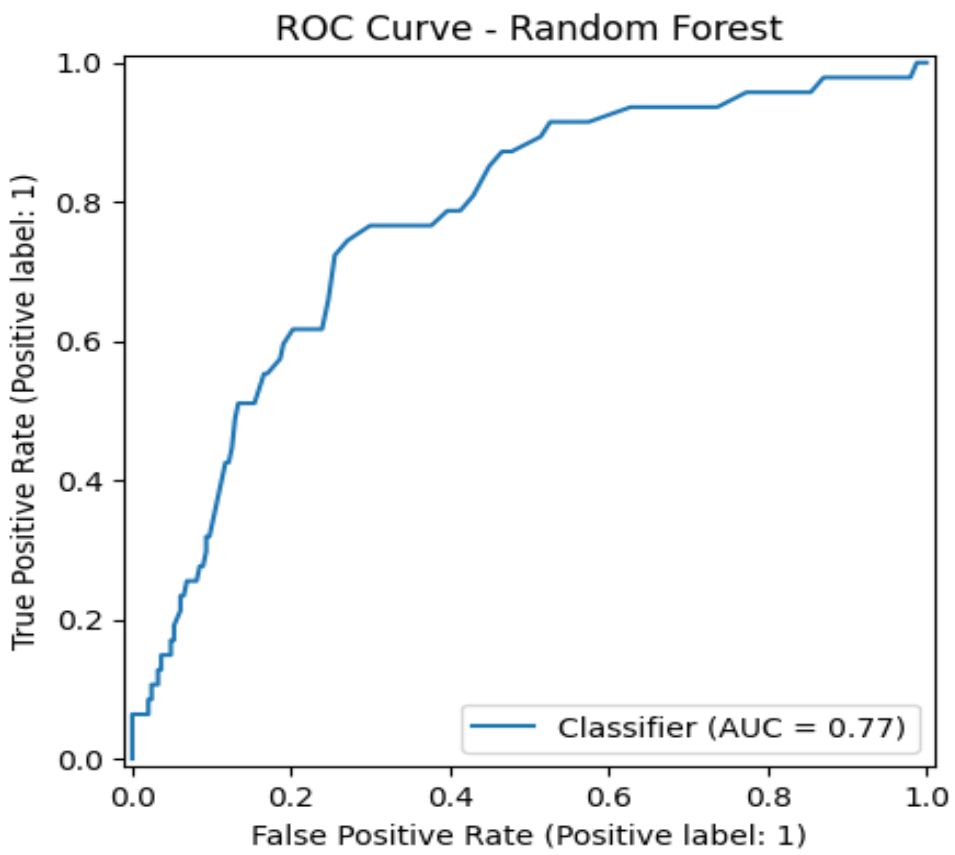
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

The most influential factors identified by the models include:

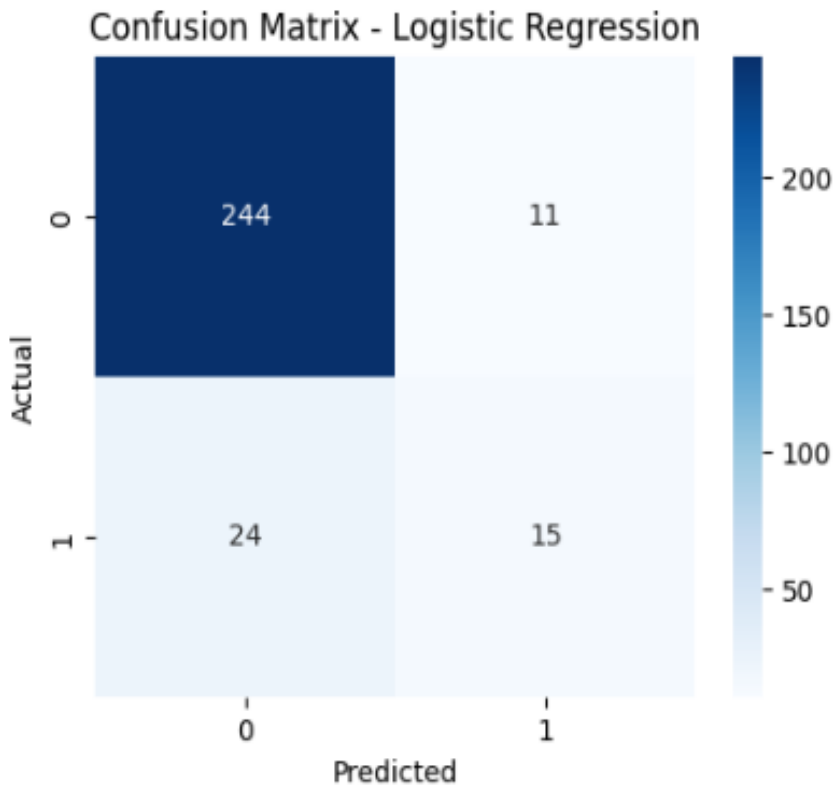
- Monthly Income
- Age
- Daily Rate
- Overtime
- Total Working Years
- Job Satisfaction
- Years at Company

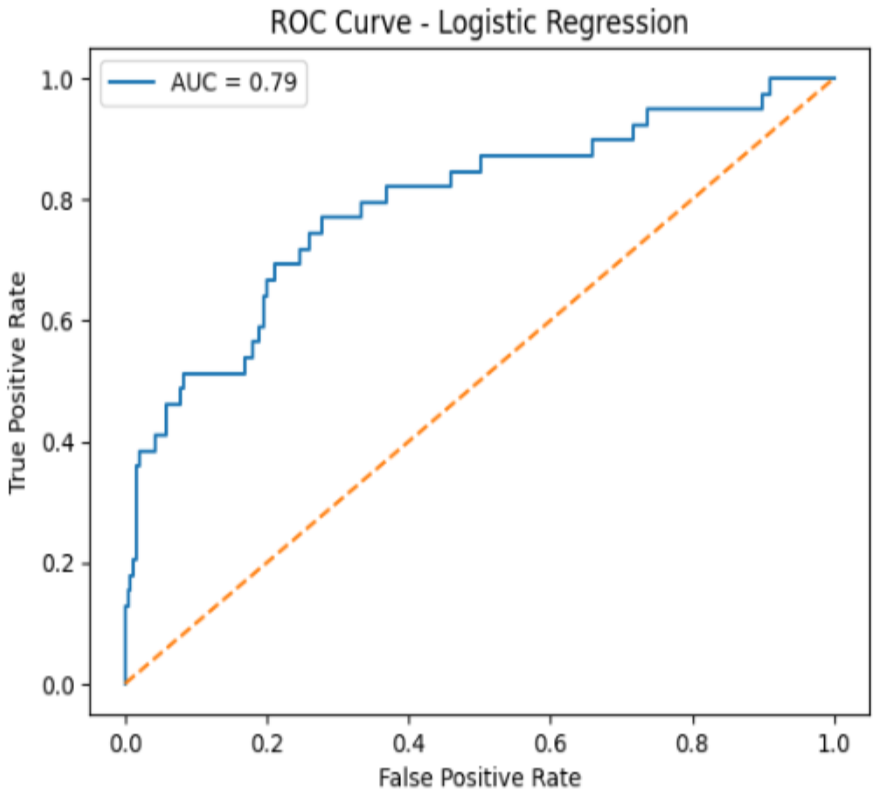
# Random Forest



# Logistic Regression



# Model Results Comparison

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.88 | 0.57 | 0.385 | 0.46 | 0.79 |
| Random Forest | 0.84 | 0.45 | 0.11 | 0.17 | 0.77 |

- Logistic Regression performed better across most evaluation metrics.
- It achieved higher accuracy, precision, recall, F1 score, and ROC-AUC.
- Random Forest struggled to detect employees who leave due to class imbalance in the dataset.

Therefore, Logistic Regression was the more reliable model for predicting employee attrition in this project.

# Cross Validation and PR Curve

5-Fold Cross-Validation Results

Logistic Regression

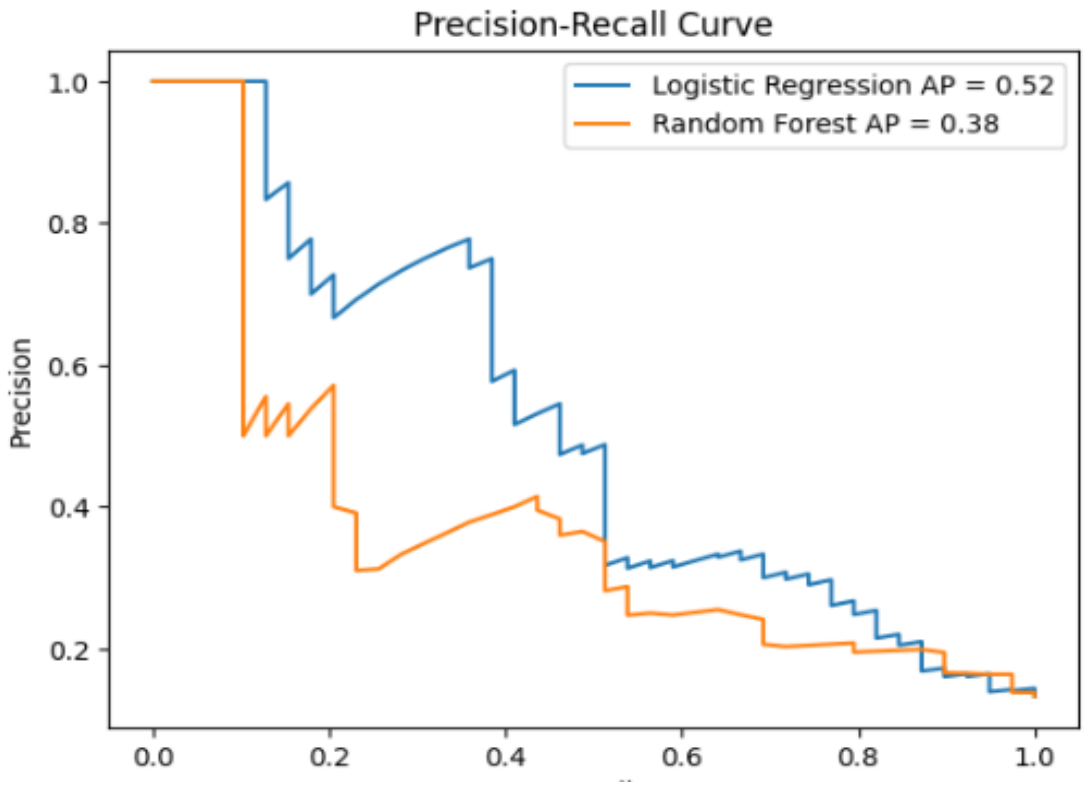
- AUC Scores (5 folds):
  - 0.858, 0.820, 0.867, 0.808, 0.865
- Mean AUC: 0.844
- Shows strong and consistent predictive performance

Random Forest

- AUC Scores (5 folds): 0.848, 0.781, 0.814, 0.773, 0.843
- Mean AUC: 0.812
- Performance is slightly lower than Logistic Regression

Key Insight

- Logistic Regression achieved higher average AUC
- Indicates better ability to distinguish employees who stay vs leave



# Challenges and Future Work

## Challenges in Predicting Employee Attrition

- **Class Imbalance:** Many more employees stayed than left → harder to predict attrition.
- **Limited Features:** Dataset lacks factors like engagement, work-life balance, leadership, and company culture.
- **Model Limitations:** Human behavior is complex; even accurate models may struggle to capture all drivers of attrition.

## Future Work: Improving Employee Attrition Prediction

- **Advanced Models:** Explore Gradient Boosting, XGBoost, or Neural Networks for better performance
- **Handle Class Imbalance:** Use techniques like SMOTE or class weighting to better predict attrition
- **Feature Engineering:** Incorporate engagement surveys, performance metrics, and work-life balance indicators.
- **Real-Time Monitoring:** Develop a system to monitor employee risk and enable preventive actions

# Conclusion

Employee Attrition Prediction: Model Comparison

- **Models Developed:** Logistic Regression & Random Forest
- **Training & Evaluation:** Same dataset used for fair comparison
- **Performance:**
  - Logistic Regression: Accuracy ≈88%, AUC 0.79
  - Random Forest: AUC ≈0.77
  - Logistic Regression performed better overall
  - Offers interpretability via coefficients → identifies key drivers (e.g., income, job satisfaction, tenure)

Logistic Regression is more suitable, balancing predictive performance and interpretability

Thank you